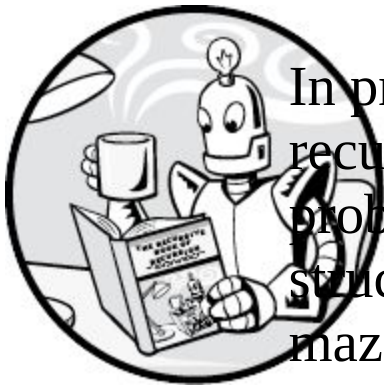


4

BACKTRACKING AND TREE TRAVERSAL ALGORITHMS



In previous chapters, you learned that recursion is especially suited for problems that involve a tree-like structure and backtracking, such as maze-solving algorithms. To see why, consider that a tree's trunk splits off into multiple branches. Those branches themselves split off into other branches. In other words, a tree has a recursive, self-similar shape.

A maze can be represented by a tree data structure, since mazes branch off into different paths, which in turn branch off into more paths. When you reach a dead end in a maze, you must backtrack to an earlier branching point.

The task of traversing tree graphs is tightly linked with many recursive algorithms, such as the maze-solving algorithm in this chapter and the maze-generation program in Chapter 11. We'll take a look at tree traversal algorithms and employ them to find certain names in a tree data structure. We'll also use tree traversal for an algorithm to obtain the deepest node in a tree. Finally, we'll see how mazes can be represented as a tree data